

Bachelor Project



**Czech
Technical
University
in Prague**

F3

**Faculty of Electrical Engineering
Department of Radioelectronics**

Design of a platform for 2D LiDAR

Illia Novosad

Supervisor: Ing. Pavel Puričér, Ph.D.

Field of study: Electronics and Communications

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Declaration

I hereby declare that artificial intelligence was used during the writing of this thesis, specifically as an aid in generating TikZ figures. Grammar was controlled with Grammarly. All generated content was verified, and I take full responsibility for the contents of this project. All information taken from the work of other authors has been properly cited.

In Prague, 13. November 2025

Abstract

A wide range of Light Detection and Ranging (LiDAR) sensors for spatial scanning is available on the market today; however, most remain too expensive for amateur use. The aim of this miniproject is to design a cost-effective platform for a 360° LiDAR sensor, which will be utilised in a subsequent bachelor's thesis to obtain a 3D model of the scanned environment. This work provides an overview of existing DIY solutions, evaluates their limitations and proposes a platform design that addresses the identified weaknesses. The result is a configurable platform that can be quickly reassembled depending on whether the user prioritises more complete coverage or quicker scan.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

Supervisor: Ing. Pavel Purič, Ph.D.

Abstrakt

Dnes je na trhu široká nabídka senzorů pro prostorové skenování; většina z nich je však příliš drahá pro amatérské využití. Cílem tohoto miniprojektu je navrhnout cenově dostupnou platformu pro 360° LiDAR senzor, která bude v navazující bakalářské práci využita k vytvoření 3D modelu naskenovaného prostředí. Práce analyzuje již existující DIY řešení, identifikuje jejich omezení a navrhuje konstrukci platformy, která tyto nedostatky omezuje. Výsledkem je modulární platforma, kterou lze rychle upravit podle požadavků uživatele.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDAR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localization and Mapping

Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [1; 2, Ch. 4-6]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the cheapest commercial 3D LiDAR sensor, which was found during the research for this miniproject, costs \$249 [3].

The primary motivation behind this mini-project (and the subsequent bachelor's thesis) is to develop a cost-effective platform for a 360° STL-19P LiDAR sensor that can be utilised for 3D spatial scanning and visualisation of the surrounding environment.



Figure 1.1: Photo of STL-19P LiDAR ¹

Table 1.1: Main parameters of STL-19P LiDAR¹

Typical measuring range [m]	0,03–12
Angular resolution [°]	0,72
Dimensions(WxLxH) [mm, mm, mm]	38,59x38,59x33,50
Communication interface	UART @ 230400 bps

A subsequent bachelor's thesis will then focus on the Digital Signal Processing (DSP) techniques for the effective processing of the acquired point cloud data and proper utilisation of open-source visualisation tools (such as Python's Open3D library [4]) to recreate a 3D scene.

¹Photo and parameters were taken from <https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/>



Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms discovered during the research for this project, analysing their pros and cons. Subsequently, the proposed design is presented, along with the chosen design decisions and their justifications.



2.1 Existing DIY Solutions for Environmental Mapping



2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilise a single-point LiDAR module mounted on a platform which can rotate and tilt. The implementation could be found in this project: [5].

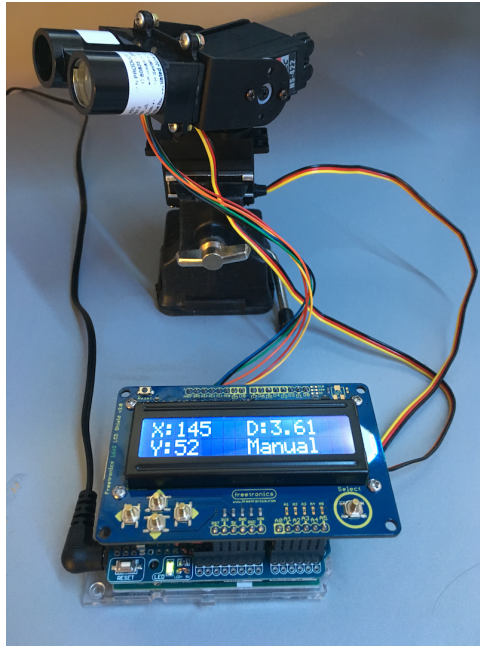


Figure 2.1: Implementation in project [5] using a single-point LiDAR

Pros

- + Single-point LiDARs are the most cost-effective among all existing types
- + Simple design

Cons

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Movement of lidar may introduce inaccuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project: [6]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360-degree sweep. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

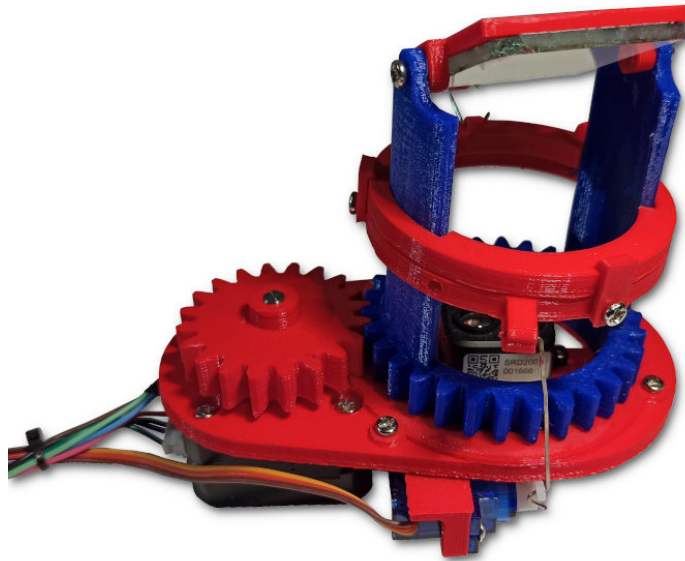


Figure 2.2: Implementation in project [6] using a single-point LiDAR and a mirror

Pros

- + Cost-effectiveness as in previous implementation
- + Simple design
- + Much easier to correct the distance measurement since the LiDAR is stationary and the distance from it to the mirror is known.

Cons

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Mirror's quality may affect the accuracy of distance measurements
- Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilises a 2D LiDAR sensor mounted on a platform that can tilt the sensor side to side, as demonstrated in this project [7]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning.

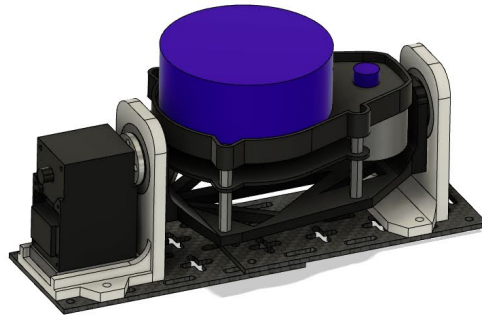


Figure 2.3: Implementation in project [7] using a tilting platform

Pros

- + Faster data acquisition compared to solutions with single-point LiDAR measurements
- + Higher resolution given by the use of a 2D LiDAR sensor
- + This design could be potentially used for Simultaneous Localization and Mapping (SLAM) applications, given the higher speed

Cons

- More expensive than single-point LiDAR solutions
- Point distribution is non-uniform due to the tilting mechanism, which produces denser point clouds on the axis of tilting (see fig. 2.4)
- Mechanical design is more prone to vibrations produced by the LiDAR itself, affecting measurement accuracy

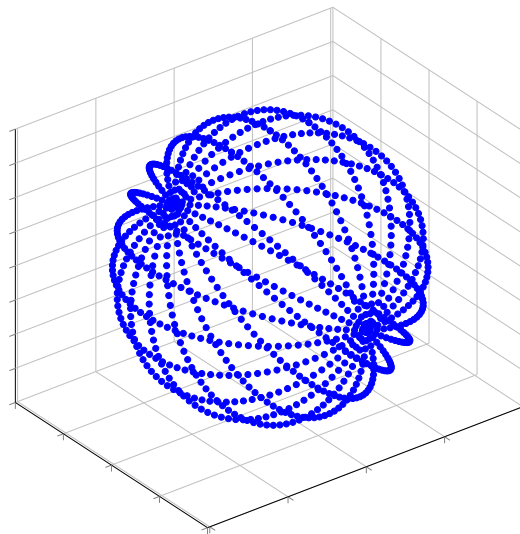


Figure 2.4: Simulation of an obtained point cloud using a tilting platform

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate around its axis.

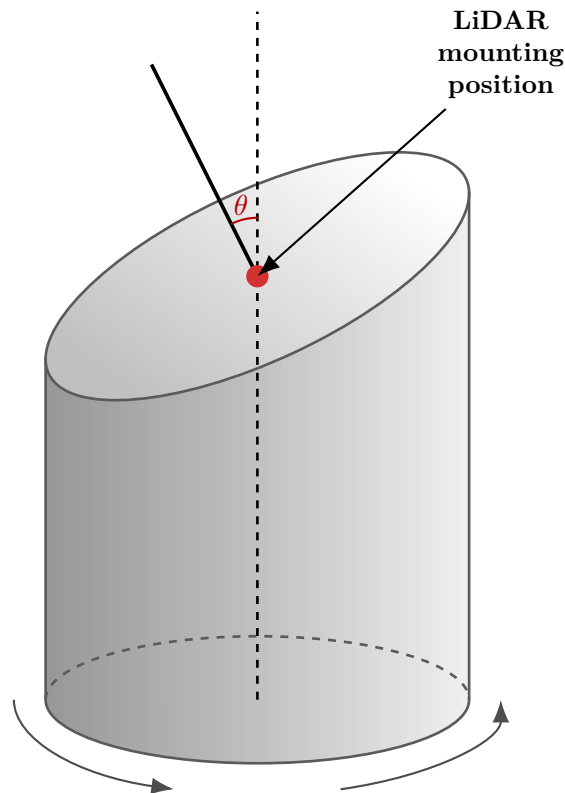


Figure 2.5: Proposed platform design. θ is the tilt angle

This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high resolution, while also eliminating some of the drawbacks of previous designs, namely:

- + The platform will be pretty rigid, minimising vibrations
- + Mechanical design is very simple, making it easy to build and troubleshoot
- + The resulting point cloud will be more uniformly distributed compared to the one obtained using the tilting platform design presented in section 2.1.3 unless the tilt angle is extremely steep (see fig. 2.6)

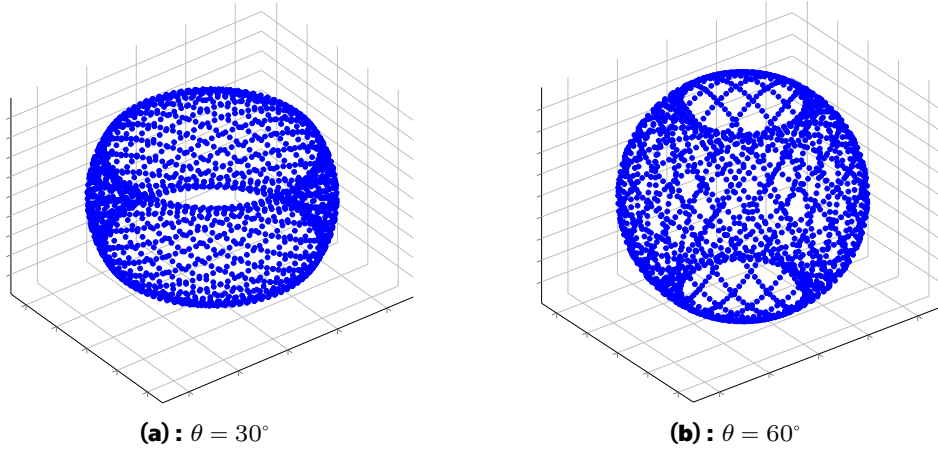


Figure 2.6: Simulation of an obtained point cloud using a rotating tilted platform with two different tilt angles θ

Such a design does have its drawbacks, however. The first one is immediately apparent from fig. 2.6: there are blind spots in the point cloud where no points are measured. While these blind spots can be minimised by choosing a large tilt angle, this approach reintroduces the problem of non-uniform point cloud density.

Besides that, compared to the tilting platform design presented in section 2.1.3, the proposed design would need to rotate across twice as large a range to obtain full coverage of the environment, since the tilting platform scanned only within $[-90^\circ, +90^\circ]$, whereas the current design must complete a full 360-degree rotation. This would lead to longer scanning times.

■ 2.2.1 Controlling the Platform

Since we want our platform to be precisely controlled, we need to choose an appropriate motor. Stepper motors are commonly used in applications requiring precise positioning, as they can be controlled to move in small, discrete steps. For this design, a NEMA 17 17HS8401 stepper motor is used to rotate the LiDAR platform.



Figure 2.7: NEMA 17 17HS8401 photo²

Table 2.1: Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor²

Motor Weight [Ω]	360
Holding Torque [N m]	0,5
Step Angle [°]	1,8
# of Steps per Revolution [-]	200
Rated Current [A]	1,68

For heavy load applications, it is important to consider the potential error sources of stepper motors, such as the stepper motor losing steps due to a heavy load or by driving the motor faster than its maximum speed [8, p. 13]. However, in our case, the load will be relatively light for this motor, and the speed will be kept low; thus, a feedback mechanism for correcting lost steps is not required.

2.2.2 Choosing the Tilt Angle

The tilt angle of the LiDAR platform is a crucial parameter that affects the size of the aforementioned blind spots in the point cloud (see fig. 2.6). The trade-off for a larger tilt angle, however, is a more spread-out point cloud

²Photo and parameters were taken from <https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/>

in the horizontal direction if we rotate our platform in discrete steps [insert figure].

As could be seen in [insert reference to figure], the resolution of the point cloud in the direction perpendicular to the internal rotation axis of the LiDAR is dependent on the tilt angle, specifically

$$\text{res} = \text{step_angle} \cdot \sin(\theta) \quad (2.1)$$

where θ is the tilt angle and step_angle is constant and is equal to 1.8° for our motor (see table 2.1). A following graph shows how the point cloud resolution changes with the tilt angle:

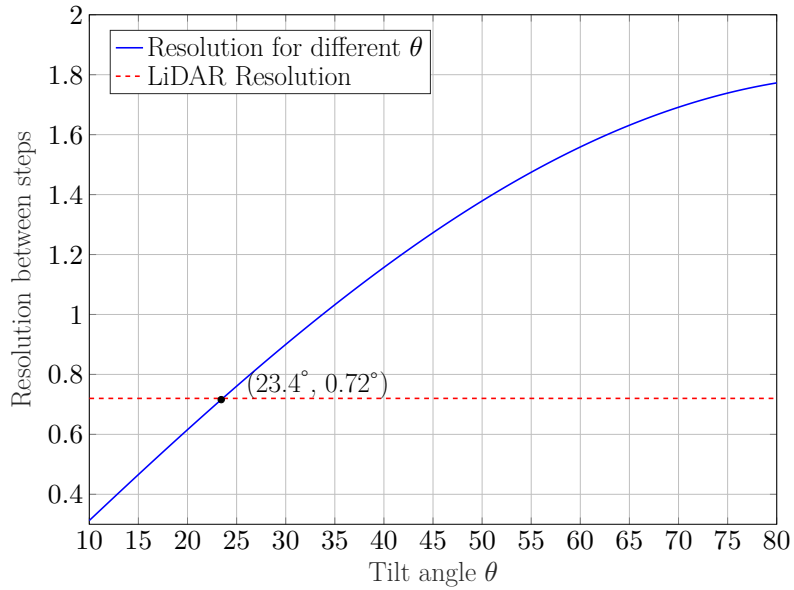


Figure 2.8: Point cloud resolution depending on the tilt angle θ

As could be seen from fig. 2.8, the resolution given by the step angle will be the same as the angular resolution of the LiDAR itself at the tilt angle of approximately 23.4° .

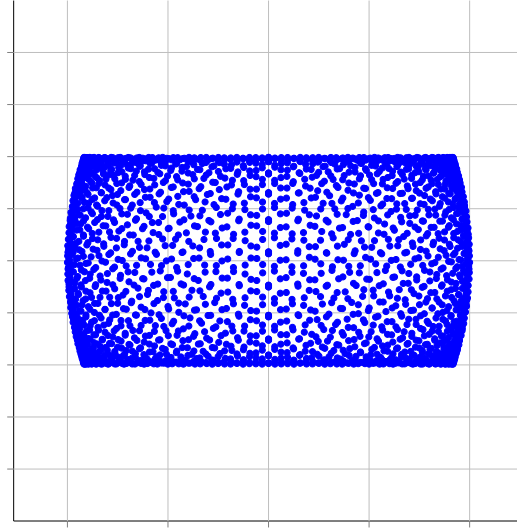


Figure 2.9: Side profile of a point cloud obtained using a rotating tilted platform with tilt angle $\theta = 23.4^\circ$

Looking at the side profile of the point cloud obtained using this tilt angle, we can see that the blind spots are extremely large making this tilt angle impractical for our application. Thankfully, there is a way to achieve a more complete coverage of the environment without sacrificing point cloud density.

2.2.3 Choosing a Gear Ratio for the Rotating Platform

In order to remove the need for a compromise between point cloud coverage and density while using a stepper motor to rotate the platform, we can introduce a gear mechanism into our design. Using gears with a specific gear ratios allows us to choose a tilt angle that minimizes blind spots while still maintaining a dense point cloud. The resolution will then be given by

$$\text{res} = \text{gear_ratio} \cdot \text{step_angle} \cdot \sin(\theta), \quad (2.2)$$

where gear_ratio is the ratio of the number of teeth on the driven gear to the number of teeth on the driving gear. A following graph is similar to fig. 2.8, but the third dimension represents different gear ratios:

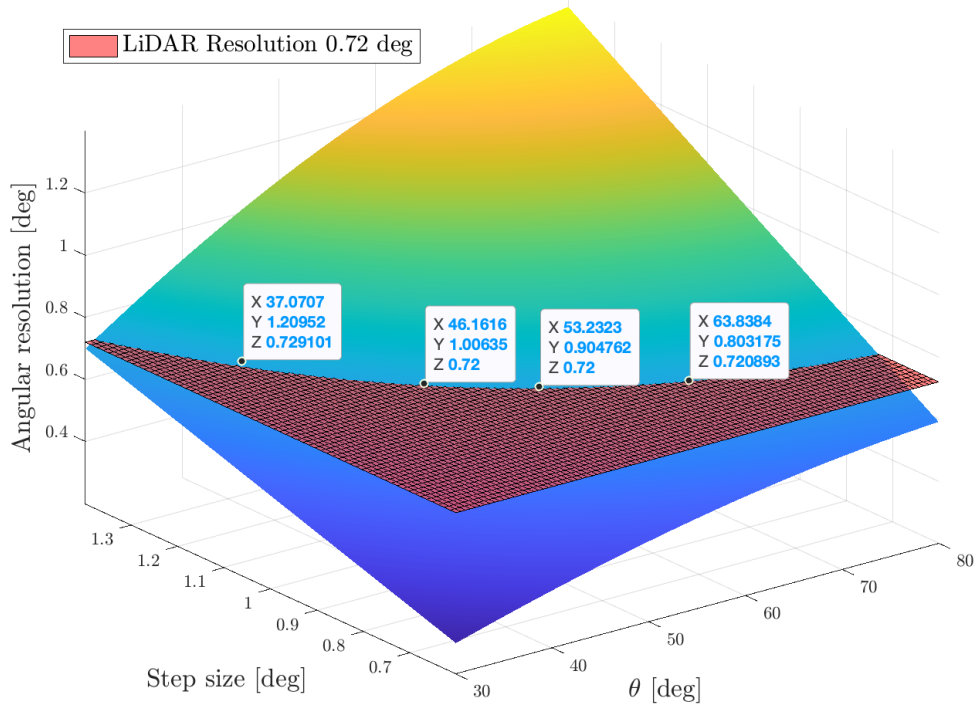


Figure 2.10: Point cloud resolution depending on the tilt angle θ and gear ratio

As could be seen from fig. 2.10, using a gearbox lets us choose from a variety of tilt angles without worsening the angular resolution.

Table 2.2: Best combinations for step angle/gear ratio/tilt angle

Step Angle [°]	Gear Ratio [-]	Tilt Angle [°]
1,2	3:2	37
1,0	9:5	46
0,9	2:1	53
0,8	9:4	64

Of course, by using the gearbox we increase the number of steps needed for the full revolution, thus increasing the scanning time. Nevertheless, by providing platforms of different sizes with appropriate gear ratios, we can make a modular platform which could be reassembled depending on the application



Chapter 3

Conclusion

The main drawbacks of the existing DIY solutions for a cheap LiDAR scanner were presented in section 2.1.

Then, the design of a rotating tilted platform was presented in section 2.2 which solved all but one problem – insufficient resolution.

Introducing a gearbox to our design is a viable solution which will not only lets us scan with higher resolution, but also choose if the scanning speed or point cloud coverage is more important in a certain scenario.

Appendix A

Bibliography

- [1] DOUBRAVA, Pavel. *LIDAR Senzor a Jeho Aplikace v Mobilním Robotu* [online]. 2021. [visited on 2025-11-23]. Available from HDL: 10467/94785. České vysoké učení technické v Praze.
- [2] DONG, Pinliang; CHEN, Qi. *LiDAR Remote Sensing and Applications*. Boca Raton: CRC Press, 2017. ISBN 978-1-351-23335-4. Available from DOI: 10.4324/9781351233354.
- [3] *4D Lidar L1 Application Scenarios_4D Lidar L1 Efficacy | Unitree Robotics* [online]. [visited on 2025-11-23]. Available from: <https://www.unitree.com/LiDAR>.
- [4] ZHOU, Qian-Yi; PARK, Jaesik; KOLTUN, Vladlen. *Open3D: A Modern Library for 3D Data Processing* [online]. 2018-01-30. [visited on 2025-10-02]. Available from DOI: 10.48550/arXiv.1801.09847.
- [5] PETERS, Dana. *LIDAR Scanner* [online]. 2022-01-19. [visited on 2025-11-17]. Available from: <https://www.qcontinuum.org/lidar-scanner>.
- [6] GRASSIN, Charles. *3D Lidar Scanner MK2* [online]. Charles' Labs, 2020-10-14. [visited on 2025-10-02]. Available from: <https://charleslabs.fr/en/project-3D+Lidar+Scanner+MK2>.
- [7] LINGKANG, Zhang. *DIY 3D Lidar* [online]. Lingkang Zhang, 2022-01-03. [visited on 2025-11-23]. Available from: <http://zlethic.com/diy-3dlidar/>.
- [8] FRIEDL, František. *Speed control stepper*. 2008. Available from HDL: 10563/6878.